Analytic Field Theory

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

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Analytic Number Theory

Throughout the readers mathematical career, they will certainly hear that complex analysis and number theory are deeply intertwined together. Any such exploration in length requires some complex analysis, at least the knowledge of factoring holomorphic functions, which is covered [1]. We shall show that every global number field shall have a Riemann-zeta function associated to it that gives information about the distribution of primes.

1.1 Riemann-Zeta Function

Definition 1.1.1: Riemann Zeta Function

The *Riemann-zeta function* is a complex function given by the series

$$\zeta_{\mathbb{Q}}(s) = \zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

and is defined on $\mathrm{Re}(s) > 1$.

Using the integral test, we see that this function converges for $\mathrm{Re}(s) > 1$ (recall the complex part only adds rotation information). Let us prove the second equality of this definition:

Proposition 1.1.2: Euler's Product

For $\mathrm{Re}(s) > 1$, we have:

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

which is absolutely convergent and $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$

***Proof*:**
We may use the layer-cake representation (see [4]), however for clarity we shall prove it more directly. We need the following chain of equalities:

$$\sum_{n=1}^{\infty} n^{-s} \overset{\text{decomp.}}{=} \sum_{n=1}^{\infty} \prod_p p^{-\nu_p(n)s} \overset{!}{=} \prod_p \left(\sum_{e \geq 0} p^{-es} \right) \overset{\text{geo. series}}{=} \prod_p \frac{1}{1-p^{-s}}$$

The key equality to show is the 2nd, which will follow by showing that the domain of convergence will be the same for all these series. Consider $\zeta_m(s)$ to be the function $\zeta(s)$ restricted to the set S_m of all integers m such that there are no prime factors $m < p$. Then there is at most $p_1, ..., p_k$ possible factors. As ζ converges absolutely on $\mathrm{Re}(s) > 1$, so does ζ_m, and so:

$$\zeta_m(s) = \sum_{m \in S_m} n^{-s} = \sum_{e_1, \ldots, e_k \geq 0} (p_1^{e_1} \cdots p_k^{e_k})^{-s} = \prod_{1 \leq i \leq k} \sum_{e_i \geq 0} (p_i^{-s})^{e_i} = \prod_{p \leq m} \frac{1}{1-p^{-s}}$$

Now, on any $\mathrm{Re}(s) > 1 + \delta$ for any delta, the sequence converges uniformly to $\zeta(s)$. Indeed, for any $\epsilon > 0$ and any s, pick appropriately small δ so that for sufficiently large m:

$$|\sigma(s) - \sigma_m(s)| \leq \left| \sum_{n \geq m} n^{-s} \right| \leq \sum_{n \geq m} |n^{-s} = \sum_{n \geq m} n^{-\mathrm{Re}(s)} \leq \int_m^{\infty} x^{-1-\delta} \leq \frac{1}{\delta} m^{-\delta} < \epsilon$$

Hence, this sequence converges locally uniformly to $\zeta(s)$ in $\mathrm{Re}(s) > 1$. By a similar argument, the sequence $P_m(s) = \prod_{p \leq m} \frac{1}{1-p^{-s}}$ converges to $\prod_p \frac{1}{1-p^s}$. To show it converges on all $\mathrm{Re}(s) > 1$, recall that a $\prod_n p_n$ converges uniformly (and absolutely) if and only if $\sum_n \log(p_n)$ converge uniformly (and absolutely). The sumation of the log converges if it's absolute value converges, hence:

$$\sum_p \left| \log\left(\frac{1}{1-p^{-s}} \right) \right| \overset{\text{Taylor Series}}{=} \sum_p \left| \sum_{e \geq 1} \frac{1}{e} |p^{-se}| \right| \leq \sum_p \sum_{e \geq 1} |p^{-s}|^e = \sum_p \frac{1}{(|p|^s - 1)} < \infty$$

this convergence is valid for $|z| < 1$ as the Taylor series of log converges on this domain. Then $\prod_p \frac{1}{1-p^{-s}}$ is absolutely convergent and nonzero on $\mathrm{Re}(s) > 1$, showing the equality and completing the proof.

1.1.3 Euler Product and Unique Factorization It was key that a number can be uniquely decomposed into a product of primes up to unit. Thus, when generalizing the Euler product to other rings R, this product exists in this exact from if R is a UFD. Otherwise, there needs to be some modifications by working with prime divisors and their norms in order to recover a product decomposition

Theorem 1.1.4: Meromorphic Extension Of Zeta Function (Analytic continuation)

$\zeta(s)$ extends meromorphically to $\mathrm{Re}(s) > 0$ with a pole at $s = 1$, namely to:

$$\zeta(s) = \frac{1}{s-1} + \varphi(s)$$

where $\varphi(s)$ is a holomorphic function on $\mathrm{Re}(s) > 0$.

***Proof*:**
let us start by showing we can write:

$$\zeta(s) = \frac{1}{s-1} + \varphi(s)$$

for a holomorphic function $\varphi(s)$ on $\mathrm{Re}(s) > 0$. To see this, take:

$$\begin{aligned}
\zeta(s) - \frac{1}{s-1} &= \sum_{n\geq 1} n^{-s} - \int_1^\infty x^{-s}dx \\
&= \sum_{n\geq 1}\left(n^{-s} - \int_n^{n+1} x^{-s}dx\right) && \text{abs. convergence} \\
&= \sum_{n\geq 1}\int_n^{n+1}(n^{-s} - x^{-s})dx
\end{aligned}$$

Define $\varphi_n(s) = \int_n^{n+1}(n^{-s} - x^{-s})dx$. Then these functions are holomorphic on $\mathrm{Re}(s) > 0$. To show $\varphi(s)$ is holomorphic, for any fixed s_0 such that $\mathrm{Re}(s_0) > 0$, and $x \in [n, n+1]$, we get:

$$\begin{aligned}
|n^{-s_0} - x^{-s_0}| &= \left|\int_n^x s_0 t^{-s_0-1}dt\right| \\
&\leq \int_n^x \frac{|s_0|}{t^{s_0+1}}dt \\
&= \int_n^x \frac{|s_0|}{t^{1+\mathrm{Re}(s_0)}}dt \\
&\leq \frac{|s_0|}{n^{1+\mathrm{Re}(s_0)}}
\end{aligned}$$

Hence,

$$|\varphi_n(s_0)| \leq \int_n^{n+1} | \leq \frac{|s_0|}{n^{1+\mathrm{Re}(s_0)}}$$

From this bound, we have that for $\epsilon = \text{Re}(s_0)/2$, on $U = B_\epsilon(s_0)$ and each $n \geq 1$:

$$\sup_{s\in U} |\varphi_n(s)| \leq \frac{|s| + \epsilon}{n^{1+\epsilon}} =: M_n$$

Then we get:

$$\sum_n M_n = (|s_0| + \epsilon)\zeta(1+\epsilon)$$

which converges. Thus, the series $\sum_n \varphi_n$ converges absolutely on $\text{Re}(s) > 0$. By the Weiestrass M-test and uniqueness of analytic extension, $\sum_n \varphi_n$ converges to a function

$$\varphi(s) = \zeta(s) - \frac{1}{s-1} \qquad \text{Re}(s) > 0$$

as we sought to show.

Corollary 1.1.5: Residue Of Riemann-Zeta Function

$$\text{Res}_{s=1}\zeta(s) = 1$$

Corollary 1.1.6: Infinite Sum of Prime Recirpicals

$$\sum_{p \text{ prime}} \frac{1}{p} \to \infty$$

***Proof*:**
By Euler's product formula:

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

Taking logarithms yields

$$\log \zeta(s) = -\sum_p \log(1 - p^{-s})$$

Expanding the logarithm using the Taylor series $\log(1-x) = -x - x^2/2 - x^3/3 - \cdots$, and using absolute convergence, we obtain:

$$\log \zeta(s) = \sum_p p^{-s} + \sum_p \sum_{k=2}^{\infty} \frac{p^{-ks}}{k}$$

The double sum converges for $s > 1/2$ since $\sum_p p^{-2s}$ converges absolutely. Hence as $s \to 1^+$,

$$\log \zeta(s) = \sum_p p^{-s} + O(1)$$

Since $\zeta(s)$ has a simple pole at $s = 1$, we have $\zeta(s) \sim 1/(s-1)$ as $s \to 1^+$, which implies $\log \zeta(s) \to \infty$. Therefore $\sum_p p^{-s}$ must diverge as $s \to 1^+$, and consequently $\sum_p 1/p = \infty$, as we sought to show.

By proposition 1.1.2, we see that $\zeta(s)$ has no roots for $\text{Re}(s) > 1$. This result can be extended to $\text{Re}(s) \geq 1$, which will be important for the prime number theorem. We require the following lemma

Lemma 1.1.7: Mertens Lemma

For any $x, y \in \mathbb{R}$ with $x > 1$

$$|\zeta(x)^3(\zeta(x+iy)^4(1z(x+2iy)| \geq 1$$

This is a "trick" that was found by Merten. For a longer proof but one with more intuition, MIT notes gives a proof by Hadamard (p.149).

***Proof*:**
As $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ for $\text{Re}(s) > 1$:

$$\log|\zeta(s)| = -\sum_p \log|1-p^{-s}| = -\sum_p \text{Re}\log(1-p^{-s}) = \sum_p \left(\sum_{n\geq 1} \frac{\text{Re}(p^{-ns}}{n}\right)$$

Pluging in $s = x + iy$, we get:

$$\log|\zeta(x+iy)| = \sum_p \left(\sum_{n\geq 1} \frac{\cos(ny\log p)}{np^{nx}}\right)$$

And so:

$$\log|\zeta(x)^3(\zeta(x+iy)^4(1z(x+2iy)| = \sum_p \sum_{n\geq 1} \frac{3 + 4\cos(ny\log p) + \cos(2ny\log p)}{np^{nx}}$$

noting that $\cos(2\theta) = 2\cos^2(\theta) - 1$, we get:

$$3 + 4\cos(\theta) + \cos(2\theta) = 2(1+\cos(\theta))^2 \geq 0$$

Taking $\theta = ny\log p$, we get:

$$\log|\zeta(x)^3(\zeta(x+iy)^4(1z(x+2iy)| \geq 0 \Rightarrow |\zeta(x)^3(\zeta(x+iy)^4(1z(x+2iy)| \geq 1$$

completing the proof.

Theorem 1.1.8: Riemann-zeta No Zeros On $x = 1$ Line

The Riemann-zeta function $\zeta(s)$ has no zeros on $\text{Re}(s) \geq 1$

***Proof*:**
By proposition 1.1.2, $\zeta(s)$ on $\mathrm{Re}(s) > 0$ has no zeros. Now, suppose that $\zeta(1+iy) = 0$ for some $y \in \mathbb{R}$. Then it must be that $y \neq 0$ as $\zeta(s)$ has a pole at $s = 1$, and $\zeta(s)$ does not have a pole at $1 + 2iy \neq 1$ by theorem 1.1.4. It follows that we must have:

$$\lim_{x \to 1} |\zeta(x^3)\zeta(x+iy)^4\zeta(x+2iy)| = 0$$

As $\zeta(s)$ has a simple pole at $s = 1$, a zero at $1 + iy$, and no pole at $1 + 2iy$. But this contradicts lemma 1.1.7, completing the proof.

In section 1.3 we shall extend this to the entire complex plane, and have an explanation for the common undergraduate "prank" of $1 + 2 + 3 + \cdots = -1/12$, which we shall see is the answer to $\zeta(-1)$. For now, let us continue on to using the zeta function for some truly incredible results.

1.2 Prime Number Theorem

Definition 1.2.1: Prime Counting Function

The *prime counting function* is:

$$\pi(x) = \sum_{p \leq x} 1$$

We shall show that:

$$\pi(x) \sim \frac{x}{\log(x)}$$

meaning that $\lim_{x\to\infty} \pi(x)/\left(\frac{x}{\log(x)}\right) = 1$. Gauss had a better approximation (meaning one whose different grows more slowly) that is asymptotically equivalent that is written down for the reader to know:

$$\pi(x) \sim \mathrm{Li}(x) = \int_2^x \frac{dt}{\log(t)}$$

We shall use the Riemann-zeta function to prove this result. This is one of the cases where there is an "elementary" way of proving this, meaning without the use of complex analysis, however the proof is much more difficult and was found around 100 years later By Erdös and Selberg in the 1940s (note that this PNT was hypothesized to be true by Gauss in the 19th century, and proven in 1896). To prove it, take the asymptotically equivalent

$$\vartheta(x) = \sum_{p \leq x} \log p$$

Lemma 1.2.2: Chebyshev's Lemma

$$\pi(x) \sim \frac{x}{\log(x)} \qquad \text{if and only if} \qquad \vartheta(x) \sim x$$

***Proof*:**
Certainly

$$0 \leq \vartheta(x) \leq \pi(x)\log(x) \qquad \Rightarrow \qquad \frac{\vartheta(x)}{x} \leq \frac{\pi(x)\log(x)}{x}$$

To bound the other side, let $\epsilon \in (0,1)$, and notice that:

$$\begin{aligned}\vartheta(x) &\geq \sum_{x^{1-\epsilon}<x\leq p} \log(p) \\ &\geq (1-\epsilon)\log(x)\left(\pi(x) - \pi(x)^{1-\epsilon}\right) \\ &\geq (1-\epsilon)\log(x)\left(\pi(x) - x^{1-\epsilon}\right)\end{aligned}$$

Thus for all ϵ,

$$\pi(x) \leq \left(\frac{1}{1-\epsilon}\right)\frac{\vartheta(x)}{\log(x)} + x^{1-\epsilon}$$

Dividing by x, we can combine the two equations to get:

$$\frac{\vartheta(x)}{x} \leq \frac{\pi(x)\log(x)}{x} \leq \left(\frac{1}{1-\epsilon}\right)\frac{\vartheta(x)}{x} + \frac{\log(x)}{x^{\epsilon}}$$

Now, the 2nd term on the right hand side tends to 0 as $x \to \infty$, and hence we can make the ratio arbitrarily close (if one tends to one, so does the other), completing the proof.

Let us start by showing that $\vartheta(x)/x$ is bounded, namely it is in $O(x)$ (finding the particular coefficient is much harder)

Lemma 1.2.3: Bounding Prime Counting Quotient

For $x \geq 1$,

$$\vartheta(x) \leq (4\log 2)(x)$$

Hence, $\vartheta(x) \in O(x)$

***Proof*:**
Take first the following:

$$2^{2n} = (1+1)^{2n} = \sum_{m=0}^{2n}\binom{2n}{m} = \frac{(2n)!}{n!n!} \overset{!}{\geq} \prod_{n<p\leq 2n} p = \exp(\vartheta(2n) - \vartheta(n))$$

where the $\overset{!}{\geq}$ inequality came from noticing $(2n)!$ is divisible for every $p \in (n, 2n]$, but $n!$ is not divisible by these p's. Taking the log of both sides, we get:

$$\vartheta(2n) - \vartheta(n) \leq 2n\log(2)$$

which is true for all $n \geq 1$. Now for any natural number $m \geq 1$,

$$\vartheta(2^m) = \sum_{n=1}^{m}\left(\vartheta(2^n) - \vartheta(2^{n-1})\right) \leq \sum_{n=1}^{m} 2^n\log(2) \leq 2^{m+1}\log(2)$$

Finally, for any real $x \geq 1$, choose an m such that $2^{m-1} \leq x < 2^m$ so that:

$$\vartheta(x) \leq \vartheta(2^m) \leq 2^{m+1}\log(2) = (4\log(2))2^{m-1} \leq (4\log(2))x$$

completing the proof.

We must now show the ratio between $\vartheta(x)$ and x is 1 asymptotically. We shall use the following elementary real analysis result to find a good criterion

Lemma 1.2.4: Integral Condition For Linear Asymptotic Equivalence

Let $f : \mathbb{R}_{\geq 1} \to \mathbb{R}$ be a non-deceasing function. Then if

$$\int_1^\infty \frac{f(t) - t}{t^2}$$

Then

$$f(x) \sim x$$

***Proof*:**

Define $F(x) = \int_1^x \frac{f(t)-t}{t^2}$. By our hypothesis $\lim_{x\to\infty} F(x)$ exists. Then for any $\epsilon > 0$, for large enough x, all the points are ϵ-close to each other. This implies that for all $\lambda > 1$ and $\epsilon > 0$:

$$|F(\lambda(x) - F(x)| < \epsilon$$

now, fix some $\lambda > 1$. Let's say that $f(x) \sim \lambda x$. Then there is an unbounded sequence λx_n such that $f(x_n) \geq \lambda x_n$ for sufficiently large n. Then:

$$F(\lambda x_n) - F(x_n) = \int_{x_n}^{\lambda x_n} \frac{f(t) - t}{t^2} \overset{!}{\geq} \int_{x_n}^{\lambda x_n} \frac{\lambda x_n - t}{t^2} dt \overset{\text{u sub.}}{=} \int_1^\lambda \frac{\lambda - t}{t^2} dt = c$$

for some constant $c > 0$, where the $\overset{!}{\geq}$ inequality comes from f being non-decreasing. Hence for large enough n,

$$|F(\lambda x_n) - F(x_n)| = c > \epsilon$$

a contradiction. It must then be that $f(x) < \lambda x$ for large enough x. A similar argument can be done to show that for sufficiently large x, $f(x) > \frac{1}{\lambda}x$, As these equality holds for all $\lambda > 1$, we get that:

$$\lim_{x\to\infty} \frac{f(x)}{x} = 1 \qquad \text{i.e.} \qquad f(x) \sim x$$

completing the proof.

To show that $\vartheta(x)$ satisfies the hypothesis of lemma 1.2.4, we shall bring in yet another real analysis tool: Take $H(t) = \vartheta(e^t)e^{-t} - 1$. Doing a change of variable argument, we see that:

$$\int_1^\infty \frac{\vartheta(t) - t}{t^2} dt < \infty \qquad \text{if and only if} \qquad \int_0^\infty H(u)du < \infty$$

The advantage of this change is that we may now use the Laplace transform to find the convergent properties. For completeness, let us box the definition of the Laplace transform:

Definition 1.2.5: Laplace Transform

Let $h : \mathbb{R}_{>0} \to \mathbb{R}$ be a piece-wise continuous function. Then the *Laplace Transform* of h, denoted $\mathcal{L}h$, is the function

$$\mathcal{L}h(s) = \int_0^\infty e^{st} h(t) dt$$

which is holomorphic for $\mathrm{Re}(s) > c$ for any $c \in \mathbb{R}$ where $h(t) = O(ce^{ct})$.

The following properties of Laplace functions should be recalled (see [3])

- linearity: $\mathcal{L}(f + ag) = \mathcal{L}(f) + a\mathcal{L}(g)$
- $g_c(x) = c$ is a constant function, then $\mathcal{L}g_c(s) = \frac{c}{s}$
- $\mathcal{L}(e^{at}h(t))(s) = \mathcal{L}(h)(s - a)$, for all $a \in \mathbb{R}$

Let us start understanding $H(u)$ after the application of the Laplace transform:

Lemma 1.2.6: Laplace Transform of Prime Counting Function

$$\mathcal{L}(\vartheta(e^t))(s) = \frac{\Phi(s)}{s}$$

is Holomorphic on $\mathrm{Re}(s) > 1$ where

$$\Phi(s) = \sum_p p^{-s} \log p$$

***Proof*:**
By lemma 1.2.3, $\vartheta(e^t) = O(e^t)$, hence $\mathcal{L}(\vartheta(e^t))$ is Holomorphic on $\mathrm{Re}(s) > 1$. Let p_n be the nth prime, and $p_0 = 0$. Then as $\vartheta(e^t)$ is constant on $t \in (\log p_n, \log p_{n+1})$, we have:

$$\int_{\log p_n}^{\log p_{n+1}} e^{st} \vartheta(e^t) dt = \vartheta(p_n) \int_{\log p_n}^{\log p_{n+1}} e^{-st} dt = \frac{1}{s} \vartheta(p_n) \left(p_n^{-s} - p_{n+1}^{-s}\right)$$

Thus:

$$\begin{aligned}(\mathcal{L}\vartheta(et))(s) &= \int_0^\infty e^{-st}\vartheta(e^t)dt \\ &= \frac{1}{s}\sum_{n=1}^\infty \vartheta(p_n)\left(p_n^{-s} - p_{n+1}^{-s}\right) \\ &= \frac{1}{s}\sum_{n=1}^\infty \vartheta(p_n)p_n^{-s} - \frac{1}{s}\sum_{n=1}^\infty \vartheta(p_{n-2})p_n^{-s} \\ &= \frac{1}{s}\sum_{n=1}^\infty \left(\vartheta(p_n) - \vartheta(p_{n-1})\right)p_n^{-s} \\ &= \frac{1}{s}\sum_{n=1}^\infty p_n^{-s}\log p_n \\ &= \frac{\Phi(s)}{s}\end{aligned}$$

as we sought to show.

Using this result, we get that:

$$\mathcal{L}H(s) = \mathcal{L}(\vartheta(e^t)e^{-t})(s) - (\mathcal{L}1)(s) = \mathcal{L}(\vartheta(e^t))(s+1) - \frac{1}{s} = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

We will show this function extends to a meromorphic function on $\mathrm{Re}(s) > 1/2$, and is holomorphic on $\mathrm{Re}(s) \geq 0$ if we eliminate the right denominators. We first require the following which finally uses the Riemann-zeta function:

Lemma 1.2.7: Meromorphic Extension of Laplace Transform

The function $\Phi(s) - \frac{1}{s-1}$ extends to a meromorphic function on $\mathrm{Re}(s) > \frac{1}{2}$ that is holomorphic on $\mathrm{Re}(s) \geq 1$

***Proof*:**
By theorem 1.1.4, $\zeta(s)$ extends to a meromorphic function on $\mathrm{Re}(s) > 0$, denote this extension by the same $\zeta(s)$. This function has a pole at $s = 1$ and no zeros on $\mathrm{Re}(s) \geq 1$. Thus, the logarithmic derivative $\frac{\zeta'(s)}{\zeta(s)}$ of $\zeta(s)$ is meromorphic on $\mathrm{Re}(s) > 0$ with no zeros on $\mathrm{Re}(s) \geq 1$ and a single simple pole at $s = 1$ with residue -1. Then by looking at the Euler product, for

$\mathrm{Re}(s) > 1$ we get:

$$\begin{aligned}
-\frac{\zeta'(s)}{\zeta(s)} &= (-\log(\zeta(s)))' \\
&= \left(-\log\prod_p \frac{1}{(1-p^{-s})}\right)' \\
&= \left(\sum_p \log(1-p^{-s})\right)' \\
&= \sum_p \frac{p^{-s}\log p}{1-p^{-s}} \\
&= \sum_p \frac{\log p}{p^s-1} \\
&= \sum_p \left(\frac{1}{p^s} + \frac{1}{p^s(p^s-1)}\right) \\
&= \Phi(s) + \sum_p \frac{\log p}{p^s(p^s-1)}
\end{aligned}$$

The sum on the right hand side will converge absolutely and locally uniformly to a holomorphic function on $\mathrm{Re}(s) > 1/2$, and the left hand side is meromorphic on $\mathrm{Re}(s) > 0$ and on $\mathrm{Re}(s) \geq 1$ it has only a simple pole at $s = 1$ with residue 1. Thus, by properties of analytic functions we get that

$$\Phi(s) - \frac{1}{s-1}$$

extends to a meromorphic function on $\mathrm{Re}(s) > \frac{1}{2}$ that is holomorphic on $\mathrm{Re}(s) \geq 1$, as we sought to show.

Corollary 1.2.8: Meromorphic Extension of $(\mathcal{L}H)(s)$

The functions $\Phi(s+1) - \frac{1}{s}$ and $(\mathcal{L}H)(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$ extend to meromorphic functions on $\mathrm{Re}(s) > -\frac{1}{2}$ that are holomorphic on $\mathrm{Re}(s) \geq 0$

***Proof*:**
The first is an immediate result of the above lemma. For the econ function, note that

$$\frac{\Phi(s+1)}{s+1} - \frac{1}{s} = \frac{2}{s+1}\left(\Phi(s+1) - \frac{1}{s}\right) - \frac{1}{s+1}$$

is meromorphic on $\mathrm{Re}(s) > -\frac{1}{2}$ and holomorphic on $\mathrm{Re}(s) \geq 0$, as it is the sum of products of such functions, completing the proof.

Theorem 1.2.9: Criterion for Laplace Transform giving Bounded Integral

Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a bounded piecewise continuous function, and suppose its Laplace transform extends to a holomorphic function $g(s)$ on $\mathrm{Re}(s) \geq 0$. Then the integral $\int_0^\infty f(t)dt$ converges and is equal to $g(0)$

***Proof*:**
Without loss of generality, assume $f(t) \leq 1$ for all $t \geq 0$. Then for $\tau \in \mathbb{R}_{>0}$, define

$$g_\tau(s) = \int_0^\tau f(t)e^{-st}dt$$

By definition

$$\int_0^\infty f(t)dt = \lim_{\tau\to\infty} g_\tau(0)$$

Hence, it suffices to show hat

$$\lim_{\tau\to\infty} g_\tau(0) = g(0)$$

To this end, pick $r > 0$, and let γ_r be the boundary of $\{s \mid |s| \leq r \text{ and } \mathrm{Re}(s) \geq -\delta_r\}$ for some chosen $\delta_r > 0$ so that g is holomorphic on γ_r (δ_r exists as g is holomorphic on $\mathrm{Re}(s) \geq 0$, and hence on some open ball $B_{\leq 2\delta(y)}(iy)$ for each $y \in [-r, r]$; take $\delta_r = \inf_{y\in[-r,r]}(\delta(y))$). Next, as each γ_r is a simple clsoed curve, and as for each $\tau > 0$ the function $h(s) = (g(s)-g_\tau(s))e^{s\tau}(1+\frac{s^2}{r^2})$ is holomorphic on a regions containing γ_r, by Cauchy's integral formula (see [1]) we can evaluate $h(0)$ to get:

$$g(0) - g_\tau(0) = h(0) = \frac{1}{2\pi i}\int_{\gamma_r} (g(s) - g_\tau(s))\, e^{s\tau}\left(\frac{1}{s} + \frac{s}{r^2}\right) ds \tag{1.1}$$

If we can show that the left hand side tends to 0 as $\tau \to \infty$, we shall get our desired result, namely that for any $\epsilon > 0$ w shall set $r = \frac{3}{\epsilon}$.

Splice $\gamma - r$ into the positive and negative semi-cirlce. Let γ_r^+ denote the positive semi-circle of radius r, i.e. restrict γ_r to $\mathrm{Re}(s) > 0$. Then as the integrand is absolutely bounded by $1/r$ on γ_r^+, for $|s| = r$ and $\mathrm{Re}(s) > 0$ we have:

$$\begin{aligned}
|g(s) - g_\tau(s)| \cdot \left|e^{s\tau}\left(\frac{1}{s} + \frac{s}{r^2}\right)\right| &= \left|\int_\tau^\infty f(t)e^{-st}dt\right| \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r} \cdot \left|\frac{r}{s} + \frac{s}{r}\right| \\
&\leq \int_\gamma^\infty e^{-\mathrm{Re}(s)t}dt \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r} \cdot \frac{2\mathrm{Re}(s)}{r} \\
&= \frac{e^{-\mathrm{Re}(s)\tau}}{\mathrm{Re}(s)} \cdot \frac{e^{\mathrm{Re}(s)\tau}}{r} \cdot \frac{2\mathrm{Re}(s)}{r} \\
&= \frac{2}{r^2}
\end{aligned}$$

Hence:

$$\left|\frac{1}{2\pi i}\int_{\gamma_r^+} (g(s) - g_\tau(s))\, e^{s\tau}\left(\frac{1}{s} + \frac{s}{r^2} ds\right)\right| \leq \frac{1}{2\pi} \cdot \pi r \cdot \frac{2}{r^2} = \frac{1}{r}$$

Now take γ_r^-, i.e. γ_r where $\mathrm{Re}(s) < 0$. Then for any fixed r, the first term $g(s)e^{s\tau}(s^{-1}sr^{-2})$ in equation (1.1) tends to 0 as $\tau \to \infty$ for $\mathrm{Re}(s) < 0$ and $|s| \leq r$. For the second term, note that as

$g_\tau(s)$ is holomorphic on $\mathbb{C}$, we may instead integrate over the semicircle of radius r in $\text{Re}(s) < 0$. For $|s| = r$ and $\text{Re}(s) < 0$, we have:

$$\begin{aligned}\left|g_\tau(s)e^{s\tau}\left(\frac{1}{s}+\frac{s}{r^2}\right)\right| &= \left|\int_\tau^\infty f(t)e^{-st}dt\right| \cdot \frac{e^{\text{Re}(s)\tau}}{r} \cdot \left|\frac{r}{s}+\frac{s}{r}\right| \\ &\leq \int_0^\tau e^{-\text{Re}(s)t}dt \cdot \frac{e^{\text{Re}(s)\tau}}{r} \cdot \frac{-2\text{Re}(s)}{r} \\ &= \left(1-\frac{e^{-\text{Re}(s)\tau}}{\text{Re}(s)}\right) \cdot \frac{e^{\text{Re}(s)\tau}}{r} \cdot \frac{-2\text{Re}(s)}{r} \\ &= \frac{2}{r^2} \cdot (1 - e^{\text{Re}(s)\tau}\text{Re}(s))\end{aligned}$$

The 2nd factor on the right hand side tends to 1 as $\tau \to \infty$ as $\text{Re}(s) < 0$. We thus get:

$$\left|\frac{1}{2\pi i}\int_{\gamma_r^-}(g(s)-g_\tau(s))\,e^{s\tau}\left(\frac{1}{s}+\frac{s}{r^2}ds\right)\right| \leq \frac{1}{2\pi}\cdot\pi r\cdot\frac{2}{r^2} \overset{\tau\to\infty}{=} \frac{1}{r}+o(1)$$

and the right hand side of equation (1.1) is bounded by $2/r + o(1)$ as $\tau \to \infty$. Hence, setting $r = 3/\epsilon$, we get for sufficiently large τ:

$$|g(0) - g_\tau(0)| < \frac{3}{r} = \epsilon$$

Hence:

$$\lim_{\tau\to\infty} g_\tau(0) = g(0)$$

as we sought to show.

Theorem 1.2.10: Prime Number Theorem

$$\pi(x) \sim \frac{x}{\log(x)}$$

***Proof*:**
Take $H(t) = \vartheta(e^t)e^{-y} - 1$. This is a piecewise continuous functions that is bounded (by lemma 1.2.2) whose Laplace transform extends to a holomorphic function on $\text{Re}(s) \geq 0$ by lemma 1.2.7. By theorem 1.2.9, the integral

$$\int_0^\infty H(t)dt = \int_0^\infty \left(\vartheta(e^t)e^{-t} - 1\right)dt$$

converges. Replacing t with $\log(x)$, we get:

$$\int_1^\infty \left(\vartheta(x)\frac{1}{x} - 1\right)\frac{dx}{x} = \int_1^\infty \frac{\vartheta(x) - x}{x^2}dx$$

converges. But then by lemma 1.2.4 $\vartheta(x) \sim x$, which by lemma 1.2.2 we get that

$$\pi(x) \sim \frac{x}{\log(x)}$$

as we sought to show.

This gives some Asymptotic behavior of the prime-counting function. Unfortunately, this does not give an error term; this is much harder to find. A proof discovered by Korobov and Vinagradov and independently has shown that this can be refined to:

$$\pi(x) = \mathrm{Li}(x) + O\left(\frac{x}{\exp\left((\log x)^{3/5+o(1)}\right)}\right)$$

with an error term $O\left(\frac{x}{(\log x)^n}\right)$ for all n, with the added restriction that the error term is not in $O(x^{1-\epsilon})$ for any $\epsilon > 0$. If we further assume the Riemann Hypothesis, which states that the zeros for $\zeta(s)$ in the critical strip $0 < \mathrm{Re}(s) < 1$ all lie on the line $\mathrm{Re}(s) = \frac{1}{2}$, then we can show that:

$$\pi(x) = \mathrm{Li}(x) + O(x^{1/2} + o(1))$$

If this could be found, then we would know a lot about the distribution of prime numbers, motivating the $1 million bounty for solving the Riemann Hypothesis.

1.3 Functional Equation

We know show how to extend $\zeta(s)$ to a meromorphic function on $\mathbb{C}$ that is holomorphic everywhere except for a simple pole at $s = 1$. Ideally, we would be able to reflect ζ across $s = 1$ and simply state:

$$\zeta(s) = \zeta(1-s)$$

However, this approach does not work, as is immediately verifiable as ζ has been extended to $s = 1/2$. Instead, an appropriate "correction factor" is introduced. In this section, we shall prove that:

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s)$$

which shall extend the Riemann-zeta function to be a meromorphic function with a single simple pole at 1. This correction factor can be thought of as considering the "real place" which is absent in the Euler product of the Riemann-zeta function.

It will quickly be noted that there are generalizations of the Riemann-zeta function which can have simpler or more complicated corrections factors; the *Dedekind-zeta functions* presented in section 1.5 will in fact be crucial in this textbook as they represent the distribution of primes in number rings, and their functional equation will be similar but more complicated. There even exist some zeta-functions (given an appropriate definition) where $\zeta(s) = \zeta(1-s)$; for example this occurs for the zeta function of elliptic curves (see [2] for the details).

1.3.1 Analysis Background his section will strongly rely on the Fourier Transform and its basic properties (including the Poisson summation formula, Plancherel's identity, its relation to convolution and differentiation, and so forth). The reader may reference [4].
Complex analysis will also be assumed, at least up to and including chapter 3 in [1].

Let us start by putting together the necessary functions and transformations.

Definition 1.3.2: Theta Function

The *theta function*[a] is defined as:

$$\Theta(\tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau}$$

[a] In [2], this function is a little more elaborate; this would be $\Theta(\tau) = \Theta_3(0; \tau)$

Observe that the sum is absolutely convergent when $\mathrm{im}\tau > 0$, and hence is holomorphic on the upper half place. Furthermore, $\Theta(\tau)$ is periodic with period 2:

$$\Theta(\tau + 2) = \Theta(\tau)$$

It further satisfies the following functional equation:

Lemma 1.3.3: Theta Function Functional Equation

For all $a \in \mathbb{R}_{>0}$,

$$\Theta(ia) = \frac{\Theta\left(\frac{i}{a}\right)}{\sqrt{a}}$$

***Proof*:**
Using Poisson summation:

$$\Theta(ia) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 a} = \sum_{n \in \mathbb{Z}} h(n) = \sum_{n \in \mathbb{Z}} \hat{h}(n) = \sum_{n \in \mathbb{Z}} g(n/\sqrt{a})/\sqrt{a} = \Theta(i/a)/\sqrt{a}.$$

Next, the Gamma function shall be recalled. First, the gamma function is a special type of transformation whose generality shall be needed later, and so is boxed here:

Definition 1.3.4: Mellin Transform

The *Mellin transform* of a function $f : \mathbb{R}_{>0} \to \mathbb{C}$ is a complex function defined by:

$$\mathcal{M}(f)(s) = \int_0^\infty f(t) t^{s-1} dt$$

From this, the Gamma function is the Mellin transformation of e^{-t}:

Definition 1.3.5: Gamma Function

$$\Gamma(s) = \mathcal{M}(e^{-t}(s) = \int_0^\infty e^{-t} t^{s-1} dt$$

The reader should recall (or prove) that $\Gamma(s)$ is holomorphic on $\mathrm{Re}(s) > 0$, and integration by parts yields the functional relation:

$$\Gamma(s+1) = s\Gamma(s)$$

for $\text{Re}(s) > 0$. Using the above equation, Γ can be extended to a meromorphic function $\mathbb{C}$ with simple poles at exactly $0, -1, -2, \ldots$. The gamma function gains its popularity by being the continuous (more specifically holomorphic) analog of the factorial of a natural number:

$$\Gamma(n+1) = n!$$

Proposition 1.3.6: Euelr's Reflection Formula

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}$$

***Proof*:**
Let $f(s) := \Gamma(s)\Gamma(1-s)\sin(\pi s)$. The function $\Gamma(s)\Gamma(1-s)$ has a simple pole at each $s \in \mathbb{Z}$ and no other poles, while $\sin(\pi s)$ has a zero at each $s \in \mathbb{Z}$ and no poles, so $f(s)$ is holomorphic on $\mathbb{C}$. Next, notice that:

$$f(s+1) = \Gamma(s+1)\Gamma(-s)\sin(\pi s + \pi) = -s\Gamma(s)\Gamma(-s)\sin(\pi s) = \Gamma(s)\Gamma(1-s)\sin(\pi s) = f(s),$$

so f is periodic with period 1. Using the substitution $u = e^t$ we get

$$|\Gamma(s)| \leq \int_0^\infty |e^{-t}t^{s-1}|dt = \int_{-\infty}^\infty |e^{-e^u}e^{u(s-1)}|e^u du = \int_{-\infty}^\infty e^{u\text{Re}(s) - e^u} du.$$

And so $|\Gamma(s)|$ is bounded on $\text{Re}(s) \in [1,2]$, hence on $\text{Re}(s) \in [0,1] \cap \text{Im}(s) \geq 1$, as $\Gamma(s+1) = s\Gamma(s)$. Then in $\text{Re}(s) \in [0,1]$:

$$|f(s)| = |\Gamma(s)||\Gamma(1-s)||\sin(\pi s)| \overset{\text{Im}(s)\to\infty}{=} O(e^{\text{Im}(s)}),$$

since $|\sin(\pi s)| = \frac{1}{2}|e^{is} - e^{-is}| = O(e^{\text{Im}(s)})$. Now, by lemma 1.3.7 (proven bellow), $f(s)$ is constant. To determine the constant, as $s \to 0$ we have $\Gamma(s) \sim \frac{1}{s}$ and $\sin(\pi s) \sim \pi s$, thus

$$f(0) = \lim_{s\to 0} f(s) = \lim_{s\to 0} \Gamma(s)\Gamma(1-s)\sin(\pi s) = \lim_{s\to 0} \frac{1}{s} \cdot 1 \cdot \pi s = \pi,$$

Uncurling, we get:

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}$$

as we sought to show.

Lemma 1.3.7: lemma – Euler's Reflection Formula

Let $f(s)$ be a holomorphic function on $\mathbb{C}$ such that $f(s+1) = f(s)$ and $|f(s)| = O(e^{\text{Im}(s)})$ as $\text{Im}(s) \to \infty$ in the vertical strip $\text{Re}(s) \in [0,1]$. Then f is constant.

***Proof*:**
The function

$$g(s) = \frac{f(s) - f(a)}{\sin(\pi(s-a))}$$

is holomorphic on $\mathbb{C}$, since $f(s) - f(a)$ is holomorphic and vanishes at the zeros $a + \mathbb{Z}$ of $\sin(\pi(s-a))$ (all of which are simple). We also have $g(s+1) = g(s)$, and $|g(s)|$ is bounded on $\mathrm{Re}(s) \in [0,1]$, since as $\mathrm{Im}(s) \to \infty$ we have $|f(s)-f(a)| = O(e^{\mathrm{Im}(s)})$ and $|\sin(\pi(s-a))| \sim e^{\pi \,\mathrm{Im}(s)}$. It follows that $g(s)$ is bounded on $\mathbb{C}$, hence constant, by Liouville's theorem. We must have $g = 0$, since

$$|g(s)| = O(e^{(1-\pi)\,\mathrm{Im}(s)}) \xrightarrow{\mathrm{im}(s)\to\infty} o(1)$$

and this implies $f(s) = f(a)$ for all $s \in \mathbb{C}$.

Using Euler's reflection formula, we can deduce ht famous identity

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

since by setting $s = 1/2$ we get $\Gamma(1/2)^2 = \pi$. More pertinently to us, we have that Γ has no zeros:

Corollary 1.3.8: Gamma Function Has No Zeroes

The gamma function $\Gamma(s)$ has no zeros on $\mathbb{C}$

***Proof*:**
For the sake of contradiction, say $\Gamma(s_0) = 0$ for some $s_0 \in \mathbb{C}$. Then note that the right hand side of the reflection formula is never zero ($\sin(\pi s)$ has no poles), and hence it must be that $\Gamma(1-s)$ has a pole at s_0. Then $1 - s_0 \in \mathbb{Z}$, so $s_0 \in \mathbb{Z}$.

Now, for $s_0 \notin \mathbb{Z}_{\leq 0}$, as Γ has poles at each negative integer and 0, and if $s_0 \in \mathbb{Z}_{>0}$ then $\Gamma(s_0) = (s_0 - 1)! \neq 0$, hence no such s_0 can exists, as we sought to show.

To conclude this section, let us take a quick overview of he infamous problem posed by the Riemann-hypothesis. First, certainly the zero's of $\xi(s)$ match the zeros of $\zeta(s)$ on the critical strip, and the zeros of $\zeta(s)$ are symmetric about the real axis (hence, it is enough to restrict our attention to the upper-half plane). Furthermore, it can be shown (and won't be here) that here are no zeros in the critical strip on the real line, hence we really need to only consider $\mathrm{im}(s) > 0$. Let $N(T)$ denote the number of zeros of $\xi(s)$ in the critical strip where $0 < \mathrm{Im}(s) < T$. For a fixed $\epsilon > 0$, we can define the rectangle:

$$R = \{-\epsilon \leq \mathrm{Re}(s) \leq 1 + \epsilon,\ 0 \leq \mathrm{Im}(s) \leq T\}$$

Then we can compute $N(T)$ via Cauchy's argument principle:

$$N(T) = \frac{1}{2\pi i} \int_{\partial R} \frac{\xi'(s)}{\xi(s)} ds$$

Given there are no zeros on the line $\mathrm{Im}9s) = T$. Then from the formula and the functional equation, we can derive:

$$N(T) \sim \frac{1}{2\pi} T \log\left(\frac{T}{2\pi e}\right)$$

In fact, we can even explicitly deduce the error term which allows us to compute $N(T)$ exactly. Already this means there are infinitely many zeros in the critical strip. As it turns out, the efforts to show that all the zeros lie on the critical line $\mathrm{Re}(s) = 1/2$ is a monumental undertaking.

One way to count the zeros on the critical line is by counting the zeros of the *Hardy Z-function*:

$$e^{i\theta(t)}\zeta(1/2+it)$$

in the region $0 \leq t \leq T$ where $\theta(t)$ is the *Riemann-Seigel function*:

$$\theta(t) = \arg\left(\Gamma\left(\frac{2it+1}{4}\right)\right) - \frac{\log(\pi)}{2}t$$

By taking asymptotic expansions of the Hardy Z-function that allow for quickly checking for its roots (including sign changes and checking for multiple roots). Then, comparing with $N(T)$, we can count whether the zeros in the critical strip in $0 < \mathrm{Im}(s0 < T$ lie on the critical line or not. So far, this has been tested for T at least 10^{13} (in other words, at least the first 10^{13} zeros above the real line all lie on the critical strip).

Functional Equation and Completing the Zeta Function

As mentioned earlier, the Riemann-zeta function does not take into account real places. The following section will present how to "correct" this and how this affect the Riemann-zeta function. First define

$$F(s) = \pi^{-s}\Gamma(s)\zeta(2s)$$

This function is holomorphic on $\mathrm{Re}(s) > 1/2$, where in this region it is absolutely convergent to the sum:

$$F(s) = \pi^{-s}\Gamma(s)\sum_{n\geq 1} n^{-2s} = \sum_{n\geq 1}(\pi n^2)^{-s}\Gamma(s) = \sum_{n\geq 1}\int_0^\infty (\pi n^2)^{-s} t^{s-1}e^{-t}dt$$

Substituting $t = \pi n^2 y$and $dt = \pi n^2 dy$, we get:

$$F(s) = \sum_{n\geq 1}\int_0^\infty (\pi n^2)^{-s}(\pi n^2 y)^{s-1}e^{-\pi n^2 y}\pi n^2 dy = \sum_{n\geq 1}\int_0^\infty y^{s-1}e^{-\pi n^2 y}dy.$$

By Fubini-Tonelli's theorem ([4, chapter 2.5]) , we can swap the sum and the integral to obtain

$$F(s) = \int_0^\infty y^{s-1}\sum_{n\geq 1} e^{-\pi n^2 y}dy.$$

Now, $\Theta(iy) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 y} = 1 + 2\sum_{n\geq 1} e^{-\pi n^2 y}$, thus

$$\begin{aligned} F(s) &= \frac{1}{2}\int_0^\infty y^{s-1}(\Theta(iy)-1)dy \\ &= \frac{1}{2}\left(\int_0^1 y^{s-1}\Theta(iy)dy - \frac{1}{s} + \int_1^\infty y^{s-1}(\Theta(iy)-1)dy\right) \end{aligned}$$

We now focus on the first integral on the RHS. The change of variable $t = \frac{1}{y}$ yields

$$\int_0^1 y^{s-1}\Theta(iy)dy = \int_\infty^1 t^{1-s}\Theta(i/t)(-t^{-2})dt = \int_1^\infty t^{-s-1}\Theta(i/t)dt.$$

By lemma 1.3.3, $\Theta(i/t) = \sqrt{t}\Theta(it)$, and adding $-\int_1^\infty t^{-s-1/2}dt + \int_1^\infty t^{-s-1/2}dt = 0$ yields

$$\begin{aligned}
&= \int_1^\infty t^{-s-1/2}(\Theta(it)-1)dt + \int_1^\infty t^{-s-1/2}dt \\
&= \int_1^\infty t^{-s-1/2}(\Theta(it)-1)dt - \frac{1}{1/2-s}.
\end{aligned}$$

Plugging this back $F(s)$ we get:

$$F(s) = \frac{1}{2}\int_1^\infty (y^{s-1}+y^{-s-1/2})(\Theta(iy)-1)dy - \frac{1}{2s} - \frac{1}{1-2s},$$

which is well-defined on $\mathrm{Re}(s) > 1/2$. Now, it is not too hard to see that

$$F(s) = F(\frac{1}{2}-s)$$

for $s \neq 0, \frac{1}{2}$, hence $F(s)$ can be analytically extend to a meromorphic function on $\mathbb{C}$ with poles only at $s = 0, 1/2$. Replacing s with $s/2$ yields the following

Definition 1.3.9: Completed Zeta Function

Let $\zeta(s)$ be the Riemann zeta function. Then the *completed Riemann Zeta Function* is:

$$Z(s) := \pi^{\frac{-s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s)$$

The completed zeta function meromorphic on $\mathbb{C}$ and satisfies the functional equation

$$Z(s) = Z(1-s)$$

As $\Gamma(s)$ has no zeros by corollary 1.3.8, the zeros of $Z(s)$ are the zeros of $\zeta(s)$. By theorem 1.1.8 and the above equation, we see that any zeros of $Z(s)$ (and $\zeta(s)$) must lie in $0 < \mathrm{Re}(s) < 1$ with one exception that will be addressed in a moment. The Riemann hypothesis claims that all these (nontrivial) zero's lie on $\mathrm{Re}(s) = 1/2$. Using the equation, it is also clear how ζ is extended as a meromorphic function non all of $\mathbb{C}$ with one simple pole at $s = 1$ (note the pole at $s = 0$ of $Z(s)$ comes from $\Gamma(s/2)$). Note that as $\Gamma(s/2)$ has poles at $-2, -4, \ldots$ while $Z(s)$ does not, we see that $\zeta(s)$ has zeros at $-2, -4, \ldots$; these are known as the *trivial zeros*.

Let us now deduce the full functional equation for the Riemann-zeta function. First, compute $\zeta(0)$ using the functional equation. Then:

$$\zeta(s) = \frac{Z(s)}{\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)} = \frac{Z(1-s)}{\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)} = \frac{\pi^{(s-1)/2}\Gamma\left(\frac{1-s}{2}\right)}{\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)}\zeta(1-s) = \frac{\pi^{s-1/2}\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)}\zeta(1-s)$$

We know that $\zeta(s)$ has a simple pole with residue 1 at $s = 1$, so

$$1 = \lim_{s\to 1^+}(s-1)\zeta(s) = \lim_{s\to 1^+}\frac{(s-1)\pi^{s-1/2}\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)}\zeta(1-s).$$

When $s = 1$, the denominator on the right hand side is $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, which cancels $\pi^{1-1/2} = \sqrt{\pi}$ in the numerator. Using $\Gamma(z) = \frac{1}{z}\Gamma(z+1)$ to shift $\Gamma\left(\frac{1-s}{2}\right)$ in the numerator yields

$$1 = \lim_{s \to 1^+} (s-1) \frac{2}{1-s} \Gamma\left(\frac{3-s}{2}\right) \zeta(1-s) = -2\Gamma(1)\zeta(0) = -2\zeta(0).$$

Thus $\zeta(0) = -1/2$. Next, Euler's reflection formula (proposition 1.3.6) to replace

$$\Gamma\left(\frac{s}{2}\right) = \pi / \left(\Gamma\left(\frac{2-s}{2}\right)\sin\left(\frac{\pi s}{2}\right)\right)$$

in the above we get:

$$\zeta(s) = \pi^{s-3/2}\Gamma\left(\frac{1-s}{2}\right)\Gamma\left(\frac{2-s}{2}\right)\sin\left(\frac{\pi s}{2}\right)\zeta(1-s).$$

Applying the *duplication formula*[1] $\Gamma(2z) = \pi^{-1/2}2^{2z-1}\Gamma(z)\Gamma\left(z + \frac{1}{2}\right)$ with $z = \frac{1-s}{2}$ then yields

Definition 1.3.10: Functional Equation of Zeta Function

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)\zeta(1-s)$$

The next object to consider is the completed zeta function with the Euler-product:

$$Z(s) = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\prod_p \left(1 - p^{-s}\right)^{-1}$$

This should be thought of a as product over the *places* of $\mathbb{Q}$, with the leading factor:

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma\left(\frac{s}{2}\right)$$

should be thought of as corresponding to the real Archimedean place of $\mathbb{Q}$. When discussing Dedekind-zeta functions, we shall have gamma factors $\Gamma_{\mathbb{R}}$ and $\Gamma_{\mathbb{C}}$ corresponding to each real/complex place.

Finally, we have the "ultimate" analytic continuation in defining the following function

Theorem 1.3.11: Holomorphic Riemann Zeta Function

The function:

$$\xi(s) = \binom{s}{2}\Gamma_{\mathbb{R}}(s)\zeta(s)$$

is holomorphic on $\mathbb{C}$ an satisfies the functional equation:

$$\xi(s) = \xi(1-s)$$

The zeros of $\xi(s)$ all lie in the critical strip $0 < \mathrm{Re}(s) < 1$.

[1]We did not prove this, and Sutherland simply links to this formula with no proof

***Proof*:**
immediate form our discussion.

This also gives the alternative functional equation for $\zeta(s)$:

$$\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \pi^{-\frac{1-s}{2}}\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s).$$

1.4 Dirichlet L-functions: Primes in Arithmetic Progressions

In this section, we shall build-up another important probabilistic fact about primes, namely that they in fact very often distribute very nicely. We build-up to the following result:

Theorem 1.4.1: Dirichlet Coprime Theorem

For all coprime integers a, m, there are infinitely many primes p such that

$$p \equiv a \bmod m$$

Furthermore, for appropriate choice of m (called the modulus), the primes shall be equi-distributed among them.

Dirichlet proved this back in 1837. However it was Riemann who showed how to establish this result using the Riemann-zeta function, and it will be a modernization of this proof that we shall present. Dirichlet's original proof was to show that:

$$\sum_{p\equiv a \bmod m} \frac{1}{p}$$

diverges. We know that $\sum_p 1/p$ diverges by corollary 1.1.6, but this is provably a smaller sum (consider $p \equiv 1 \bmod 4$). The idea then is to "extract" the primes that satisfy this relationship and somehow show that they also have a divergent sum. The extraction will be done via *Dirichlet Characters* and the new functions based off of a character will be called an L-function.

We shall require the following results later in the proof:

Theorem 1.4.2: Mertens Theorem

as $x \to \infty$:

1. $\sum_{p\leq x} \frac{\log p}{p} = \log(x) + R(x0$ where $|R(x)| < 2$
2. $\sum_{p\leq x} = \log\log x + B + O\left(\frac{1}{\log x}\right)$ where $B = 0.2514..$ is *Mertens' constant*
3. $\sum_{p\leq x} \log\left(1 - \frac{1}{p}\right) = -\log\log x - \gamma + O\left(\frac{1}{\log x}\right)$ where $\gamma = 0.57721..$ is *Euler's Constant*

***Proof*:**
exercise.

1.4.1 Dirichlet Characters and L-Functions

The following results are certainly familiar to the reader as they show up in the representation theory of finite groups. Dirichlet's theorem came about before group theory was common use:

Definition 1.4.3: Arithmetic Function

A function $f : \mathbb{Z} \to \mathbb{C}$ is called an *arithmetic function*. The function f is *multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all coprime $m, n \in \mathbb{Z}$. It is totally multiplicative if $f(1) = 1$ and $f(mn) = f(m)f(n)$. For $m \in \mathbb{N}_{>0}$, we say that f is *m-periodic* if

$$f(n+m) = f(n) \qquad n \in \mathbb{Z}$$

and m is called the period of f if it is the least $m > 0$ for which this is satisfied .

Definition 1.4.4: Dirichlet Character

A *Dirichlet character* is a periodic totally multiplicative arithmetic function, and is usually denoted $\chi : \mathbb{Z} \to \mathbb{C}$.

The Dirichlet Character is almost a group homomorphism. An m-periodic Dirichlet character must map 1 to 1, $f(1+m)$ to 1, and satisfy $f(nm) = f(n)f(m)$. The reader can check that such ha function can always be restricted to $(\mathbb{Z}/m\mathbb{Z})^*$, which now from a group character. Hence, it's image must be a finite subgroup of $U(1)$. Conversely, we can start with a group character on $(\mathbb{Z}/m\mathbb{Z})^*$ and extend to $\mathbb{Z}$ by mapping the rest of the elements to 0. In fact, there is a correspondence between Dirichlet characters and group-characters where a group character $\chi : (\mathbb{Z}/m\mathbb{Z})^* \to \mathbb{C}$ can be extended to $\mathbb{Z}$ by $\chi(n) = 0$ whenever $\gcd(n, m) = 1$ (or equivalent notation: $n \not| m$). The constant $\mathbb{1}(n) = 1$ is the *trivial Dirichlet character*, and is the unique Dirichlet character with period 1.

Definition 1.4.5: Dirichlet Character of Modulus m

Let χ be Dirichlet character. Then it is said to be *of modulus* m if it is an m-periodic Dirichlet character that is the extension of a group character on $(\mathbb{Z}/m\mathbb{Z})^\times$.

Notice that being m-periodic isn't sufficient to have a unique extension, for example a 2-periodic Dirichlet character can have modulus 2^k character.

Example 1.4.6: Dirichlet Characters

1. The character on $(\mathbb{Z}/2\mathbb{Z})^*$ has a unique extension, but this character is also the trivial character on $\mathbb{Z}/4\mathbb{Z}^* = \{1, 3\}$ where $\chi(1) = \chi(3) = 1$. Similarly for each 2^k $k \geq 1$.
2. The key way to think of Dirichlet characters modulo m is that they remember multiplicity information with the character information. Let $e \in \hat{G}$ be a character of G. Then we can think of a Dirichlet Character as:

$$\chi(n) = \begin{cases} e(n) & \gcd(n,m) = 1 \\ 0 & \text{otherwise} \end{cases}$$

   This also motivates the need for irreducible Dirichlet characters, which shall be introduced momentarily.

In general, we have the following

Lemma 1.4.7: Inducing same Dirichlet Character

Let χ be a Dirichlet character with period m. Then χ is a Dirichlet character of modulus m' if and only if $m|m'|m^k$ for some k.

***Proof*:**
Let us show that $\chi(n) \neq 0$ if and only if $m \perp n$. To that end, suppose $\chi(n) \neq 0$ but $m \not\perp n$. Let p be a common divisor f m and n. Then $\chi(p) \neq 0$. As $\chi(p)\chi(n/p) = \chi(n) \neq 0$ we have that for any $r \in \mathbb{Z}$:

$$\chi(r)\chi(p) = \chi(rp) = \chi(rp + m) = \chi(r + m/p)\chi(p)$$

hence $\chi(r) = \chi(r+m/p)$ (as $\chi(p) \neq 0$. And so this implies χ is (m/p)-periodic, which contradicts the minimality of m. Hence $\chi(n) \neq 0 \Rightarrow m \perp n$.

Next, for any $n \perp m$, pick $a = ne \equiv 1 \mod m$, giving

$$\chi(1) = \chi(a) = \chi(n^e) = \chi(n)^e \neq 0$$

so $\chi(n) \neq 0$, and so $n \perp m \Rightarrow \chi(n) \neq 0$, so χ is a Dirichlet character of modulus m,

Finally, if $m|m'|mk$, then the prime divisors of m' coincide with those of m, and so :

$$n \perp m' \Leftrightarrow n \perp m \Leftrightarrow \chi(n) \neq 0$$

Certainly χ is m'-periodic (as $m|m'$ so χ is a Dirichlet character of modulus m', Conversely, if χ is a Dirichlet character of modulus m' ,then χ is a m'-periodic, and thus $m|m'$, (since m is he period of χ). As χ is a Dirichlet character of modulus m and of modulus m', for each prime p we have:

$$p|m \Leftrightarrow \chi(p) = 0 \Leftrightarrow p|m'$$

and so the prime divisors of m and m' coincide, and so m' divide some power of m for the right k, completing the proof.

This motivates the idea of a minimal or primitive Dirichlet character

Definition 1.4.8: Primitive Dirichlet Character

Let χ_1, χ_2 be Dirichlet character of modulus m_1, m_2 respectively where $m_1 \big| m_2$. Then if $\chi_2(n) = \chi_1(n)$ for $n \in (\mathbb{Z}/m_2\mathbb{Z})^\times$, we say χ_2 is *induced* by χ_1. If a Dirichlet character is induced by no other character, it is said to be *primitive.*

Lemma 1.4.9: Induced Dirichlet Characters

A Dirichlet character χ_2 of modulus m_2 is induced by a Dirichlet character of modulus $m_1 \big| m_2$ if and only if χ_2 is constant on residue classes in $(\mathbb{Z}/m_2\mathbb{Z})^\times$ that are congruent modulo m_1. If this is the case, the Dirichlet character χ_1 of modulus m_1 that induces χ_2 is uniquely determined

***Proof*:**
exercise (very doable, but if stuck look at MIT Chap 18 page 4)

Definition 1.4.10: Principal Dirichlet Character

Let χ be a Dirichlet character. Then χ is called *principal* if it is induced by $\mathbb{1}$. Let $\mathbb{1}_m$ denote the principal Dirichlet character of moduli m corresponding toe the trivial character on $(\mathbb{Z}/m\mathbb{Z})^\times$

Note that a principal primitive character is necessarily $\chi = \mathbb{1}$. The principal character $\mathbb{1}_m$ is given by:

$$\mathbb{1}_m(n) = \begin{cases} 1 & \gcd(n,m) = 1 \\ 0 & \gcd(n,m) > 1 \end{cases}$$

Lemma 1.4.11: Characterizing Principal Characters

Let χ be a Dirichlet character of modulus m. Then:

$$\sum_{n \in \mathbb{Z}/m\mathbb{Z}} \chi(n) \neq 0 \qquad \Leftrightarrow \qquad \chi = \mathbb{1}_m$$

***Proof*:**
This is a Representation Theory argument. We have that $\chi(n) = 0$ for $n \notin (\mathbb{Z}/m\mathbb{Z})^\times$ if and only if χ restricts to the trivial character on $(\mathbb{Z}/m\mathbb{Z})^\times$. (see [5, chapter 24.2]).

Theorem 1.4.12: Characterizing Dirichlet Characters

Every Dirichlet character χ is induced by a primitive Dirichlet character $\tilde{\chi}$ that is uniquely determined by χ

***Proof*:**
exercise (also very doable, but more book keeping. If you're stuck see MIT notes Chapter 18 p.4)

Definition 1.4.13: Conductor of Dirichlet Character

The *conductor* of a Dirichlet character χ is the period of the unique primitive Dirichlet character $\tilde{\chi}$ that induces χ

Corollary 1.4.14: Characterizing conductor 1 Character

For any Dirichlet character χ of modulus m we have

$$\sum_{n \in \mathbb{Z}/m\mathbb{Z}} \chi(n) \neq 0 \qquad \text{if and only if} \qquad \chi \text{ has conductor } 1$$

***Proof*:**
follow directly from lemma 1.4.11

Corollary 1.4.15: Relating Dirichlet Character to all Concepts

Let $M(m)$ denote the set of Dirichlet characters of modulus m, and $X(m)$ denote the set of primitive Dirichlet characters of conductor dividing m, and $\widehat{G}(m)$ denote the character group of $(\mathbb{Z}/m\mathbb{Z})^\times$. hen we have canonical bijections:

$$M(m) \to X(m) \to \widehat{G}(m) \qquad \chi \mapsto \widehat{\chi} \mapsto (n \mapsto \widehat{\chi}(n))$$

***Proof*:**
exercise (if stuck MIT chap 18 p 5, but really should be something to do on your own)

We can now make $X(m)$ into a group via $\tilde{\chi_1}\tilde{\chi_2} = \widetilde{\chi_1\chi_2}$. The result is not given by pointwise multiplication of the two separate characters, but the unique primitive character that induces the pointwise product.

Example 1.4.16: 12-periodic Dirichlet Character
In the following, the period is m and the conductor is c.

m	c	0	1	2	3	4	5	6	7	8	9	10	11	mod-12	principal	primitive
1	1	1	1	1	1	1	1	1	1	1	1	1	1	no	yes	yes
2	1	0	1	0	1	0	1	0	1	0	1	0	1	no	yes	no
3	1	0	1	1	0	1	1	0	1	1	0	1	1	no	yes	no
3	3	0	1	-1	0	1	-1	0	1	-1	0	1	-1	no	no	yes
4	4	0	1	0	-1	0	1	0	-1	0	1	0	-1	no	no	yes
6	1	0	1	0	0	0	1	0	1	0	0	0	1	yes	yes	no
6	3	0	1	0	0	0	-1	0	1	0	0	0	-1	yes	no	no
12	4	0	1	0	0	0	1	0	-1	0	0	0	-1	yes	no	no
12	12	0	1	0	0	0	-1	0	-1	0	0	0	1	yes	no	yes

Note that the exponent of $(\mathbb{Z}/m\mathbb{Z})^\times$ is 2, and so $\chi(n) \in \{0, -1, 1\}$ as $(\text{im}\chi) \cap U(1) \subseteq \mu_2 = \{\pm 1\}$.

We shall now look at the distributions of primes using the following modified Riemann-Zeta function:

Definition 1.4.17: Dirichlet L-Function

Let χ be a Dirichlet Character. Then the *Dirichlet L-function* associated to χ is:

$$L(s,\chi) = \prod_p (1 - \chi(p)p^{-s})^{-1} = \sum_{n\geq 1} \chi(n) n^{-s}$$

Naturally, the product converges absolutely for $\text{Re}(s) > 1$ (as $|\chi(n)| \leq 1$, and hence $L(s,\chi)$ is holomorphic on $\text{Re}(s) > 1$). If $\chi = \mathbb{1}$, Then $L(s, \mathbb{1}) = \zeta(s)$. The key realisation is that for each principle character $\mathbb{1}_m$ of modulus m we have we can re-write η as:

$$\zeta(s) = L(s, \mathbb{1}_m) \prod_{p \mid m} (1 - p^{-s})^{-1}$$

Notice that the product term on the right hand side is finite (even though $\mathbb{1}_m$ is zero for infinitely many numbers, they are all *not* coprime to m). Hence, $L(s, \mathbb{1}_m$ has a simple pole at $s = 1$ with residue:

$$\mathrm{Res}_{s=1}(L, \mathbb{1}_m) = \lim_{s\to 1^+}(s-1)\zeta(s)\prod_{p|m}(1-p^{-s}) = \prod_{p|m}(1-p^{-1}) = \frac{\varphi(m)}{m}$$

where $\varphi(m)$ is the Euler totient. Note that the L-function of a non-principal Dirichlet character does *not* have a pole at $s = 1$:

Proposition 1.4.18: L-function Holomorphic

Let χ be a non-principal Dirichlet character of modulus m. Then $L(s, \chi)$ extends to a holomorphic function on $\mathrm{Re}(s) > 0$.

Note that this can be extended to when χ is simply non-principal, but the added complexity won't justify its addition.

***Proof*:**
Define the function $T : \mathbb{R}_{\geq 0} \to \mathbb{C}$ by

$$T(x) := \sum_{0<n\leq x} \chi(n).$$

By corollary 1.4.14, for any $x \in \mathbb{R}_{\geq 0}$

$$T(x+m) - T(x) = \sum_{x<n\leq x+m} \chi(n) = \sum_{n\in\mathbb{Z}/m\mathbb{Z}} \chi(n) = 0,$$

since χ is non-principal. Thus $T(x)$ is periodic modulo m and therefore bounded.

Writing $L(s, \chi)$ as a Stieltjes integral (see [4] or [6]) and integrating by parts yields

$$\begin{aligned}
L(s,\chi) &= \sum_{n\geq 1}\chi(n)n^{-s} \\
&= \int_0^\infty x^{-s}dT(x) \\
&= x^{-s}T(x)\big|_0^\infty - \int_0^\infty T(x)d(x^{-s}) \\
&= 0 - \int_0^\infty T(x)(-sx^{-s-1})dx \\
&= s\int_0^\infty T(x)x^{-s-1}dx.
\end{aligned}$$

The right hand side is holomorphic on $\mathrm{Re}(s) > 0$, since it is the limit of the uniformly converging sequence of functions $\varphi_n(s) := s\int_0^n T(x)x^{-s-1}dx$ as $T(x)$ is bounded, and is thus the analytic continuation of $L(x, \chi)$ to $\mathrm{Re}(s) > 0$.

1.4.2 Final step for Dirichlet Theorem

From this, we can now come back to proving that

$$\sum_{p \equiv a \bmod m} p^{-s}$$

diverges as $s \to 1^+$. By using the inner product on characters, the sum can be converted to:

$$\frac{1}{\varphi(m)} \sum_{\chi \in X(m)} \chi\left(\frac{p}{a}\right) = \begin{cases} 1 & p \equiv a \bmod m \\ 0 & \text{otherwise} \end{cases}$$

Where p/a is computed modulo m, and χ ranges over the primitive Dirichlet characters of conductor dividing m (which can be identified with the characters of the group $(\mathbb{Z}/m\mathbb{Z})^*$). Thus, as $s \to 1^+$:

$$\begin{aligned}
\sum_{p \equiv a \bmod m} p^{-s} &= \sum_p p^{-s} \frac{1}{\varphi(m)} \sum_{\chi \in X(m)} \chi(p/a) \\
&= \sum_{\chi \in X(m)} \frac{\chi(1/a)}{\varphi(m)} \sum_p \chi(p) p^{-s} \\
&= \sum_{\chi \in X(m)} \frac{\chi(1/a)}{\varphi(m)} (\log L(s, \chi) + O(1)) \\
&= \frac{\log \zeta(s)}{\varphi(m)} + \sum_{\substack{\chi \in X(m) \\ \chi \neq \mathbb{1}}} \frac{\chi(1/a)}{\varphi(m)} \log L(s, \chi) + O(1).
\end{aligned}$$

Now, *if* $L(1, \chi) \neq 0$ when χ is not principal, then $\log(L(s, \chi)) = O(1)$ as $s \to 1^+$ and:

$$\sum_{p \equiv a \bmod m} p^{-s} = \frac{\log \zeta(s)}{\varphi(m)} + O(1)$$

showing the sum is unbounded since $\zeta(1)$ is unbounded by corollary 1.1.6. Finally, by applying Mertens Theorem (theorem 1.4.2):

$$\sum_{\substack{p \leq x \\ p \equiv a \bmod m}} \frac{1}{p} \sim \frac{\log \log x}{\varphi(m)}$$

showing there are infinitely many primes where $p \equiv a \bmod m$. We shall prove the key *if* in the next section by proving a slightly more general result that will apply to number rings more generally.

The ideas presented here took a specific subset of primes. We shall sometimes want to study different subsets of primes and see their properties. The following shall be useful:

Definition 1.4.19: Dirichlet Density

The *Dirichlet density* of a set of primes S is given by:

$$d(S) = \lim_{s \to 1^+} \frac{\sum_{p \in S} p^{-s}}{\sum_p p^{-s}}$$

defined whenever the limit exists.

It should be noted that there is a other common density that shall be more prevelant in section **??**:

Definition 1.4.20: Natural Density

The *natural density* of a set of primes S is given by:

$$d(S) = \lim_{x \to \infty} \frac{\#\{p \le x \mid p \in S\}}{\#\{p \le x\}}$$

defined whenever the limit exists.

From the above deduction ,we can find that the Dirichlet density of $S = \{p \equiv a \bmod m\}$ is

$$d(S) = \lim_{s \to 1^+} \frac{\sum_{p \in S} p^{-s}}{\sum_p p^{-s}} = \lim_{s \to 1^+} \frac{\log \zeta(s)/\varphi(m)}{\log \zeta(s)} = \frac{1}{\varphi(m)}$$

This brings us to the amazing conclusion that the primes modulo m are equidistributed! In particular, for all $a \perp m$:

$$\sum_{\substack{p \le x \\ p \equiv a \bmod m}} \frac{1}{p} \sim \frac{1}{\varphi(m)} \sum_{p \le m} \frac{1}{p} \sim \frac{1}{\varphi(m)} \log \log x$$

It must be noted that we can in fact do better by combining this with a slight generization of the prime number formula. Let $\varphi(x; m, a)$ equal the number of primes $p \le x$ where $p \equiv a \bmod m$. Then:

$$\pi(x; m, a) \sim \frac{1}{\varphi(m)} \pi(x)$$

1.5 Dedekind-Zeta Functions

It was mentioned that the key missing fact to complete the proof of Dirichlet's theorem is that the L-function $L(s, \chi)$ is holomorphic and non-vanishing at $s = 1$ for all non-principal Dirichlet characters χ. The proof of this can be subsumed into a more general proof when we consider the Riemann zeta function over number fields more generally.

As number fields are no longer unique factorization domains, the Euler product formula would not work. We thus "recover" unique factorization by instead taking the (absolute) norm of ideals:

Definition 1.5.1: Dedekind Zeta Functions

Let K be a number field. Then

$$\zeta_K(s) = \sum_{\mathfrak{a}} \mathrm{Nm}(\mathfrak{a})^{-s} = \sum_{I} \mathfrak{N}(\mathfrak{a})^{-s} = \prod_{p \neq 0} (1 - \mathfrak{N}(\mathfrak{p})^{-s})^{-1}$$

Lemma 1.5.2: Relating Zeta Functions

For $\sigma \in \mathbb{R}_{>1}$, $d = [K : \mathbb{Q}]$, $\zeta_K(\sigma)$ converges absolutely and is $\leq \zeta(\sigma)^d$

***Proof*:**

For each prime number p, factor $p\mathcal{O}_K = \prod_{i=1}^{g} p_i^{e_i}$, $\mathrm{Nm}(P_i) = p_i^{f_i}$, $\sum_i^g e_i f_i = d$. Then the "p-factor of $\zeta_K(\sigma)$" is

$$\prod_{i}^{g} (1 + p^{-f_i \sigma} + p^{-2f_i \sigma} + \cdots) \leq (1 + p^{-\sigma} + p^{-2\sigma} + \cdots)^d$$

with equality if p splits completely

We can now ask for the residue of ζ_K at 1. Recall that $\mathrm{Res}_{s=1}\zeta(s) = 1$. For a number ring more generally, the value is a good deal more complicated. In this section, we endeavour to prove the following:

Theorem 1.5.3: Analytic Class Number Formula

$\zeta_k(s)$ extends meromorphically to $s > 1 - \frac{1}{d}$ with a unique pole (which is simple) at $s = 1$ with residue

$$\mathrm{Res}_{z=1}\zeta_k(z) = \frac{2^{\pi}(2^r(2\pi)^s h_K R_K}{w_K |D_K|^{1/2}}$$

where:

- r, s are the number of real/complex embeddings respectively
- $h_K = \#\mathrm{Cl}\,(\mathcal{O}_K)$ is the class number
- R_K is the *regulator*
- $w_K = \#\mu_K$ is the number of roots of unity
- $D_K = \mathrm{disc}(\mathcal{O}_K)$ is the absolute discriminant.

Remember that$|D_K|^{1/2}$ is the covolume of $\mathcal{O}_K$ and R_K i the covolume of $\Lambda_K = \mathrm{Log}(\mathcal{O}_K^\times)$ (see [5]).

The name of the theorem comes from the class number is in general the hardest number to calculate in this equation. In fact, the class number is often estimated by understanding with enough precision the rest of the terms in this equation. Note this proof has nothing to do with the zeta function. We shall show another theorem and show it implies this one

Example 1.5.4: Analytic class formula for Rationals

We should insure that we get $\text{Res}_{z=1}\zeta_{\mathbb{Q}}(z) = 1$. In this case:

$$n = 1,\ r = 1,\ s = 0,\ h = 1,\ w = \#\{\pm 1\} = 2,\ D = 1$$

The regulator R is the covolume of a lattice in a zero-dimension vector space, hence the determinant as a 0×0 matrix, which tautologically is 1. Thus:

$$\lim_{z \to 1^+} (z-1)\zeta_{\mathbb{Q}}(z) = \frac{2^1(2\pi)^0 1 \cdot 1}{2|1|^{1/2}} = 1$$

Before proving this result, it is worth-while showing for a moment how this ties with natural density

Theorem 1.5.5: Equivalent Theorem – Natural Density

Define $N_K(x) = \#\{I \subseteq \mathcal{O}_K \mid \text{Nm}(I) \leq x\}$ (notice that $N_{\mathbb{Q}}(x) = \lfloor x \rfloor$), and let A_k be the residue of ζ_K at 1. Then

$$N_K(x) = A_K x + O(x^{1-\frac{1}{d}})$$

where O is big-O notation.

***Proof*:**

Thm 2 imply Thm 1

$$\zeta_k(s) = \sum N(I)^{-s}$$

$$\sum n^{-s}(N_K(n) - N_K(n-1))$$

$$\begin{aligned} &= \sum N_k(n)(n^{-s} - (n_1)^{-s}) \\ &= \sum (A_k n + O(x^{1-1/d}))(n^{-s} - (n+1)^{-s}) \\ &= \sum O(n^{-1/d - \text{Re}(s)}) \end{aligned}$$

where for the last line not that $O(n^{-\text{Re}(s)-2}) = n^{-s} - (n+1)^{-s}$. This converges absolutely for $\text{Re}(s) > 1 - \frac{1}{d}$. So

$$\sum A_k n(n^{-s}(n+1)^{-s}) = A_k \zeta(s)$$

as we sought to show.

Example 1.5.6: Application

1. Let $K = \mathbb{Q}(i)$. Then $\mathcal{O}_K \setminus \{0\} \xrightarrow{4-1} \{\text{non-zero ideals}\}$ given by $\alpha \mapsto (\alpha)$. Then

$$\begin{aligned} N_K(x) &= \frac{1}{4}\#\{\alpha \in \mathcal{O}_K \mid N(\alpha) \leq x\} \\ &= \frac{-1}{4} + \frac{1}{3}\#\left\{(a,b) \in \mathbb{Z}^2 \;\middle|\; a^2 + b^2 \leq x\right\} \\ &= \frac{1}{3}\pi x + O(x^{1/2}) \end{aligned}$$

where we get the πx estimate by counting lattice points within a circle or radius $\sqrt{x}$, giving us $\sim \pi x$.

2. Let $K = \mathbb{Q}(\sqrt{2})$. Then

$$\mathcal{O}_K \setminus \{0\} \xrightarrow{\infty - 1} \{\text{non-zero ideals}\}$$

For simplicity, we may quotient by $\mathcal{O}_K$ to get an isomorphism.

1.5.1 Lipschitz parameterizability

here MIT 19.1

1.5.2 Proof of Analytic Class Number Formula

Theorem 1.5.7: Bound on Ideals

Let K be a number field of degree n. As $t \to \infty$, the number of nonzero $\mathcal{O}_K$-ideals $\mathfrak{a}$ of absolute norm $\mathrm{N}(\mathfrak{a}) \leq t$ is

$$\left(\frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}}\right) t + O\left(t^{1-1/n}\right),$$

where r and s are the number of real and complex places of K, respectively, $h_K = \#\mathrm{cl}\,\mathcal{O}_K$ is the class number, R_K is the regulator, $w_K := \#\mu_K$ is the number of roots of unity, and $D_K := \operatorname{disc} \mathcal{O}_K$ is the absolute discriminant.

***Proof*:**
MIT 19.2

Theorem 1.5.8: Analytic Class Number Formula Proof

Let K be a number field of degree n. The Dedekind zeta function $\zeta_K(z)$ extends to a meromorphic function on $\mathrm{Re}(z) > 1 - \frac{1}{n}$ that is holomorphic except for a simple pole at $z = 1$ with residue

$$\lim_{z \to 1^+} (z-1)\zeta_K(z) = \rho_K := \frac{2^r (2\pi)^s h_K R_K}{w_K |D_K|^{1/2}},$$

where

1. r and s are the number of real and complex places of K
2. $h_K = \#\mathrm{cl}\,\mathcal{O}_K$ is the class number
3. R_K is the regulator
4. $w_K = \#\mu_K$ is the number of roots of unity
5. $D_K = \operatorname{disc} \mathcal{O}_K$ is the absolute discriminant.

***Proof*:**
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1.5.3 Proof of non-vanishing L-Function

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